

Mathematics PYQ-Pattern Practice

Original exam-pattern practice; not verbatim past-paper copies

Algebra & Number-Based Practice

- Solve a quadratic and classify its roots using the discriminant.
- Given two roots of a quadratic, construct the equation satisfying those roots.
- Simplify a complex-number expression and find its modulus.
- Find a requested term of an AP or GP from given conditions.

Matrices & Coordinate Geometry

- Find the inverse of a 2×2 matrix and use it to solve two simultaneous equations.
- Find the equation of a line through two given points.
- Determine the centre and radius of a circle from its general equation.
- Use a determinant to test collinearity of three points.

Calculus

- Evaluate a standard limit after algebraic simplification.
- Differentiate a composite function using the chain rule.
- Find intervals of increase/decrease of a function.
- Determine local maxima/minima from derivative information.
- Evaluate a definite integral using a property or substitution.

Vectors, 3D & Probability

- Find the angle between two vectors using their dot product.
- Find the area of a parallelogram from two vectors.
- Determine whether two 3D lines are parallel/perpendicular from their direction data.
- Calculate a conditional probability from a two-stage experiment.
- Find mean/variance from a small discrete data set.

Answer Guidance

- For quadratic questions, start with $D=b^2-4ac$.
- For matrix questions, check determinant before attempting an inverse.
- For calculus questions, simplify first and then choose the derivative/integration rule.
- For probability, define events and sample space before calculating.